Q-1:

Cache index = 129 mod 4

= 1000 0001 mod 2^2

= 01 (Ans)

Q2

block address = $\left\lfloor \frac{321}{10} \right\rfloor = \lfloor 32.1 \rfloor = 32 = 10\ 0000$ (Ans)

Q3

Block Size = 32 bits = 4 bytes

Cache Size = 64 bits = 8 bytes = $\frac{8}{4}$ = 2 blocks

Block Add. = $\left\lfloor \frac{230}{4} \right\rfloor = \lfloor 59.75 \rfloor = 59$

Index = 59 mod 2

= 1 (Ans)

Q-4

Block Size = 8 bits = 1 byte

Cache Size = 32 bits = 4 bytes = $\frac{4}{1}$ = 4 blocks

Index bit size = $\log_2(4)$ = 2 bits

Offset " " = $\log_2(1)$ = 0 bits

Tag " " = (9 - 2 - 0) = 7 bits.

Address = 297

= 100101001

9 bits

Address = 100101001

M L

Tag bits | Index bits

Index bits = 01

Tag bits = 1001010

No offset bits (Ans)

Q-5

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
010	1	011	Data
011	1	001	Data
100	0		
101	1	000	Data
110	1	010	Data
111	0		

from the table, block size = 8 bits = 1 byte

Cache " = 8 blocks

∴ Index size = $\log_2(8)$ = 3 bits

∴ Offset " = $\log_2(1)$ = 0 "

∴ Tag " = (6 - 3) = 3 bits

a)

	Mem. Add.	hit / miss
1	100 100 (T I)	miss
2	111 010 (T I)	miss
3	010 110 (T I)	hit
4	100 100 (T I)	hit
5	111 110 (T I)	miss

b) Hit Ratio = $\frac{2}{5}$

Miss " = $\frac{3}{5}$

c) AMAT = Hit time + Miss Rate × Miss Penalty

= $2 + \frac{3}{5} \times 5 = 5$

d) 2 → 1 cycle

1 → $\frac{1}{2}$ "

∴ 5 → $\frac{5}{2}$ " = 2.5 cycle.

So, 2.5 cycles are required per instruction

Updated cache table after access 1 –

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
010	1	011	Data
011	1	001	Data
100	~~0~~ 1	100	~~Data~~
101	1	000	Data
110	1	010	Data
111	0		

Updated cache table after access 2 –

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
010	~~0~~ 1	~~011~~ 111	~~Data~~ Data
011	1	001	Data
100	~~0~~ 1	100	Data
101	1	000	Data
110	1	010	Data
111	0		

As, access-3 was a hit, so cache table remains the same.

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
010	~~0~~ 1	~~011~~ 111	~~Data~~ Data
011	1	001	Data
100	~~0~~ 1	100	Data
101	1	000	Data
110	1	010	Data
111	0		

As, access-4 was a hit, so cache table remains the same.

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
010	~~0~~ 1	~~011~~ 111	~~Data~~ Data
011	1	001	Data
100	~~0~~ 1	100	Data
101	1	000	Data
110	1	010	Data
111	0		

Updated cache table after access 5 –

Cache table:

Index	Valid bit	Tag	Data(8 bits)
000	1	100	Data
001	0		
010	~~0~~ 1	~~011~~ 111	~~Data~~ Data
011	1	001	Data
100	~~0~~ 1	100	Data
101	1	000	Data
110	~~0~~ 1	~~010~~ 111	~~Data~~ Data
111	0		